Visual Document Understanding for Hierarchical Table Extraction and Business Insights

Advanced Computer Vision for Deep Learning

Final Project Report

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1. Abstract

Complex business documents such as financial reports, procurement forms, and operational schedules often contain tables with multi-level headers, merged cells, and irregular layouts. Traditional OCR systems can extract text from these documents, but they usually do not preserve the logical structure of the table. As a result, the extracted output is difficult to use for downstream analytics, reporting, or business decision-making.

This project develops a visual document understanding pipeline for hierarchical table extraction and structured data generation. The pipeline is organized around two tasks: Table Structure Recognition (TSR), which converts document table images into HTML-style table structures, and Key Information Extraction (KIE), which converts the reconstructed table output into analytics-ready business records. The modeling section evaluates both trainable models and external baselines, including UniTable, Florence-2, SPARTAN, LlamaParse, and ChatGPT, using the FinTabNet dataset and a stratified sample of wide, tall, low-contrast, low-quality, and normal table images.

Model performance is evaluated with structure-aware metrics such as Skeleton Tree-Edit-Distance-based Similarity (TEDS) and Pipeline TEDS, along with KIE and end-to-end usability measures. Current results show that structure reconstruction and full table reconstruction are not equally difficult: UniTable achieves a strong Skeleton TEDS score of 0.8353, but its Pipeline TEDS remains much lower after content integration, even after fine-tuning. This gap suggests that cascading errors between structure prediction, cell localization, and content insertion remain a major challenge. To support downstream use, the project also includes an HTML-to-DataFrame parsing layer that converts model-generated HTML into rectangular tables for export, KIE, and dashboard readiness checks.

2. Exploratory Data Analysis

2.1 EDA Objective and Data Scope

This study uses the FinTabNet dataset, a large-scale benchmark for table structure recognition in financial documents. FinTabNet contains diverse table layouts with different levels of structural complexity and image quality, making it suitable for evaluating document understanding models on realistic financial tables.

For the EDA, we used the processed FinTabNet-derived hierarchical table dataset prepared during the initial data preparation stage of the project. Each record links a table image id with its ground-truth HTML table and the features extracted from both the image and the HTML structure.

The goal of this analysis is to understand where table complexity comes from, including merged cells, wide or tall layouts, and image quality issues such as blur or low contrast. These findings are later used to guide sampling and model evaluation.

EDA item	Value	Interpretation
Total records	32,670	Full processed feature table used for EDA
Tables with colspan > 1	32,043 (98.1%)	Horizontal header/cell spanning, common in multi-level financial tables
Tables with rowspan > 1	2,137 (6.5%)	Less common, but important for row hierarchy
Image-level missing values	1 row	Negligible missingness in image measurements

2.2 Schema and Feature Engineering

The combined EDA file contains 21 columns. The HTML-derived features describe table structure, including row count, cell count, maximum colspan, maximum rowspan, and span flags. The image-derived features include width, height, aspect ratio, blur score, contrast score, and image_type.
A simple hard_score was also created to identify difficult examples. It is not a model metric. It only helps rank tables by structural complexity and size, so that hard cases can be included in later evaluation.

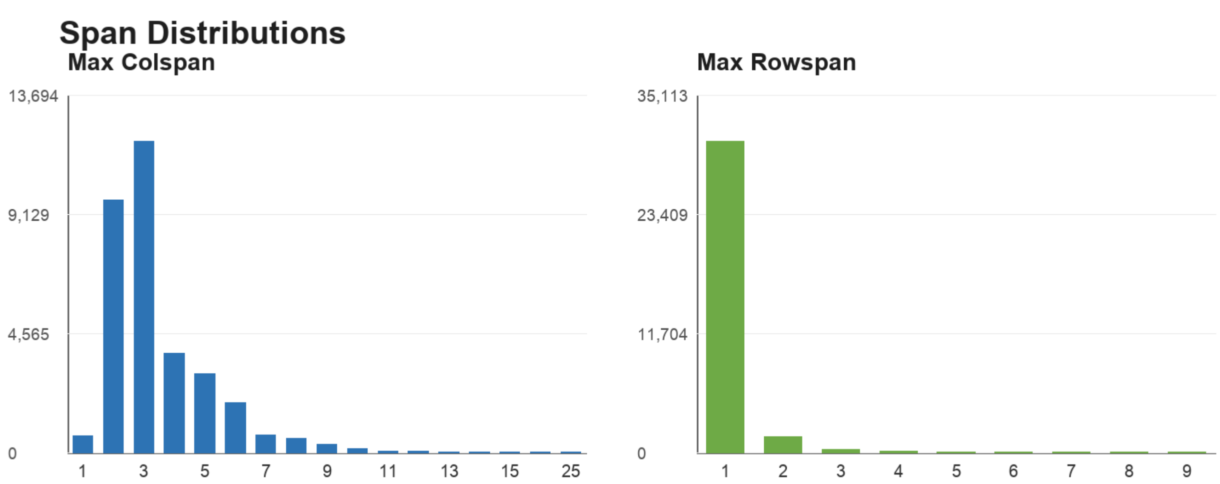
Feature	Observed range	EDA interpretation
num_rows	2 to 83	Mean 15.1; tall tables create long decoding sequences
num_cells	4 to 963	Mean 75.1; dense tables amplify alignment errors
max_colspan	1 to 25	Median 3; horizontal header grouping is common
max_rowspan	1 to 9	Median 1; vertical span is rarer but still critical

width / height	506-2085 px / 78-2477 px	Widths are relatively stable, while heights vary strongly
aspect_ratio	0.38 to 21.43	Median 2.93; wide tables are a major visual pattern
blur_score	153.8 to 17782.6	Most samples are not severely blurred
contrast_score	17.8 to 114.6	Low contrast exists but is not the dominant challenge

2.3 Univariate Analysis of Structure

Horizontal spanning is the clearest structural pattern. Only 627 tables have max_colspan = 1, so 98.1% of the records contain at least one horizontally merged cell. Most max_colspan values are 2 or 3, which suggests that grouped headers are common in this dataset.

Vertical spanning is much less frequent. About 93.5% of records have max_rowspan = 1, while 6.5% contain at least one vertically merged cell. This smaller subset still matters because rowspan errors can shift later cells and damage the table hierarchy.



Distributions of maximum colspan and maximum rowspan across the processed table records.

2.4 Image-Level EDA

Image width is fairly stable, usually around 1,400-1,600 pixels, but height varies much more. This creates two different layout problems: very wide tables with compressed columns, and tall tables with many rows. Wide tables are the largest group at 48.6% of the records, followed by normal tables at 34.8% and tall tables at 9.7%.

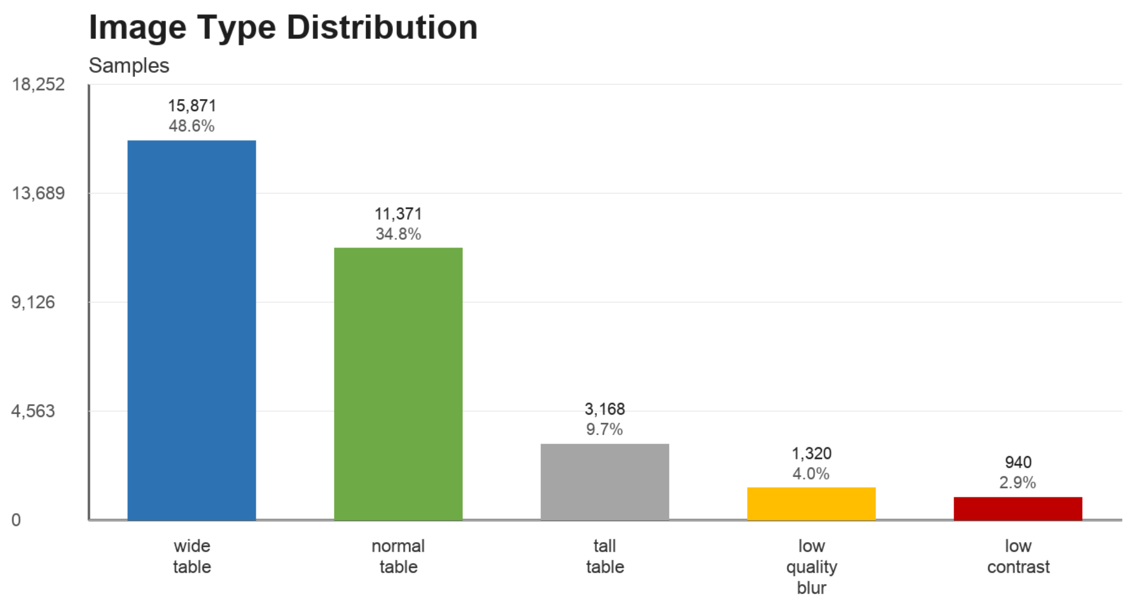
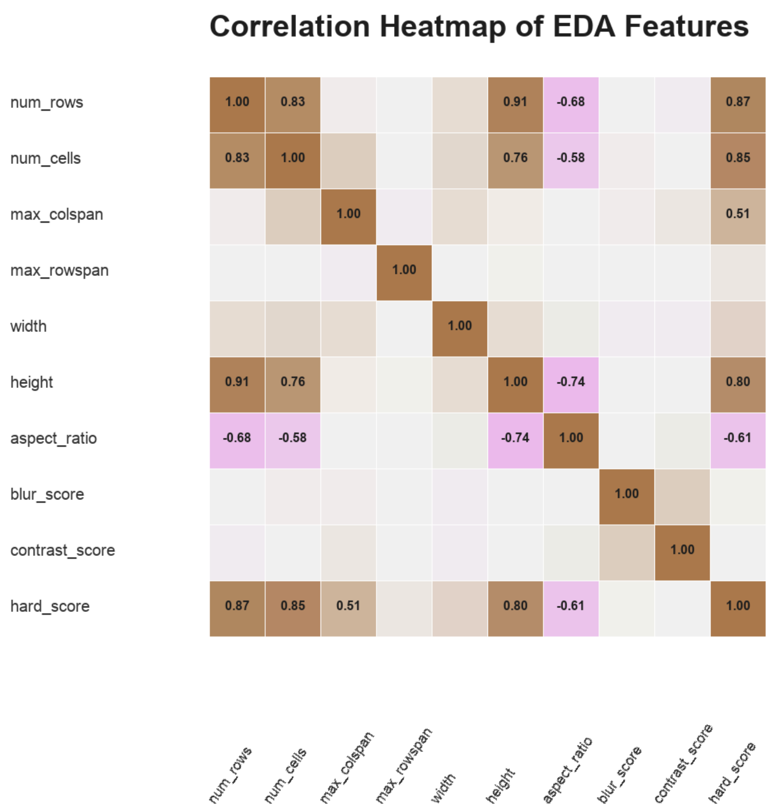


Image category distribution derived from width, height, aspect ratio, blur score, and contrast score.

Blur and contrast were checked as image-quality factors, but they are not the main source of difficulty. Most images are clear enough, and nearly 80% of contrast scores fall between 40 and 60. The harder cases are mainly caused by layout and hierarchy rather than unreadable images.

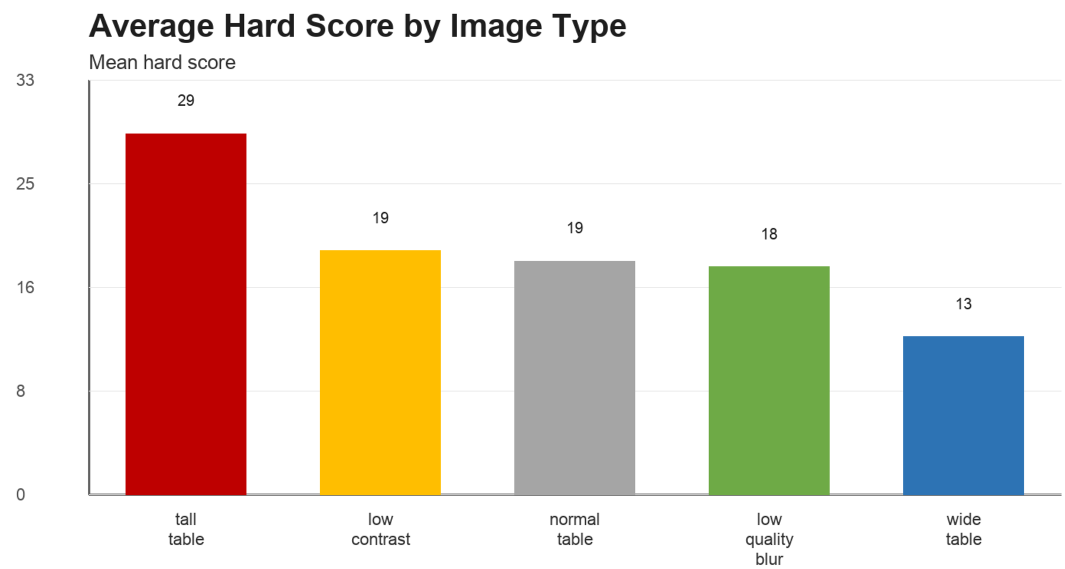
2.5 Bivariate Analysis and Hard-Example Selection

The feature relationships point in the same direction. Row count is strongly correlated with image height ($r = 0.91$) and hard_score ($r = 0.87$), while blur and contrast have weak relationships with hard_score. This again suggests that layout complexity explains more of the difficulty than image quality.

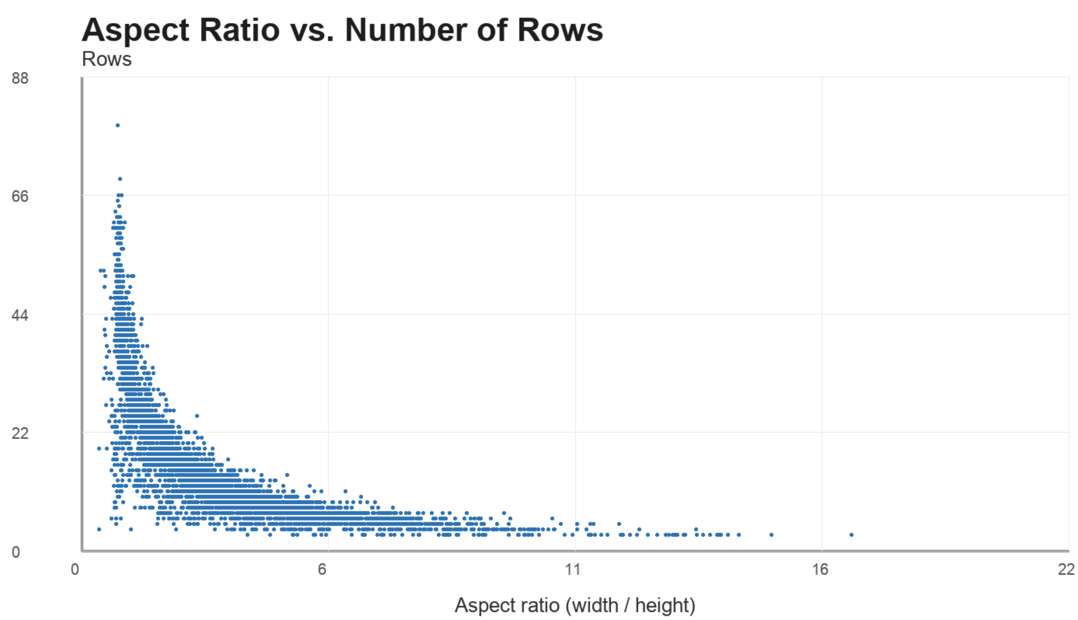


Correlation heatmap for structural, visual, and difficulty features.

Tall tables have the highest average hard_score because they contain many rows and cells. Wide tables score lower on this heuristic, but they are still important because resizing can compress columns and make header-to-column alignment fragile.



Average hard_score by image type; tall tables are the highest-scoring stress cases.



Aspect ratio versus number of rows shows the separation between wide and tall table challenges.

2.6 Implications for Model Evaluation

These findings explain why the later evaluation should include hard examples instead of only normal tables. The dataset contains many merged headers and many wide layouts, so the model should be judged on whether it preserves structure, not only whether it extracts text.

This also supports the use of Skeleton TEDS and Pipeline TEDS. TEDS compares HTML tree structure, so it is better aligned with the project goal than plain text accuracy alone. The full feature table was used for EDA, while a smaller stratified subset was later used for model training and evaluation.

The TEDS validation checks behaved as expected: identical HTML produced TEDS = 1.0, clearly different HTML produced TEDS = 0, and a partially similar pair produced TEDS = 0.404. These results confirm that the metric captures structural similarity between HTML tables.

2.7 EDA Findings

- The dataset is structurally complex, especially because horizontal spanning is very common.
- Rowspan appears less often, but it is important because it affects row hierarchy.
- The main visual challenge is wide and tall table layouts. Their impact may not be limited to skeleton structure recognition, but may also increase the difficulty of full table reconstruction.
- Hard examples and structure-aware metrics are necessary for a fair model evaluation.

3. Model Training and Evaluation

3.1 Problem Formulation

This study addresses the problem of converting complex document table images into structured, machine-readable representations for downstream analytical applications. The proposed framework consists of two primary tasks: TSR and KIE.

- **Table Structure Recognition (TSR):** TSR focuses on reconstructing the logical structure of a table from a document image by generating a structured HTML representation that preserves rows, columns, merged cells, and hierarchical relationships.

- **Key Information Extraction (KIE):** KIE focuses on extracting semantically meaningful information from the reconstructed table to produce analytics-ready business records.
- **End-to-End Pipeline Objective:** The proposed pipeline transforms unstructured table images into structured representations through the following workflow:

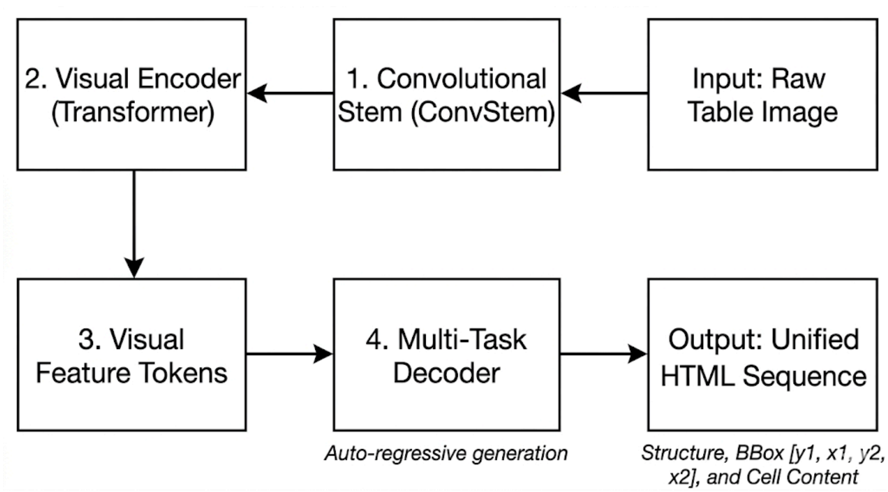
  Document Image → TSR → HTML → KIE → Structured Business Records

3.2 Model Architectures

This study evaluates both trainable models and external API-based systems for table structure recognition (TSR) and end-to-end document understanding. Specifically, UniTable, Florence-2, and Pytesseract are trained or fine-tuned models, while LlamaParse and ChatGPT are used as external API baselines. This design enables comparison across task-specific models, general multimodal models, and production-grade systems.

3.2.1 UniTable

UniTable is a foundation model for TSR that employs a pixel-to-text encoder-decoder architecture. While traditional vision transformers that convert images directly into patches perform poorly due to the physical contradiction between the receptive field (RF) ratio and sequence length, where larger patch sizes expand the RF but shorten the feature sequence length, thereby limiting the transformer's ability to store complex contextual relationships, UniTable utilizes a hybrid visual encoder where a convolutional stem (ConvStem) is combined with a Transformer. This ConvStem optimizes the receptive field independently of sequence length, enabling the model to efficiently capture both local geometric details and global layouts directly from raw images. Additionally, during its self-supervised pre-training stage, UniTable introduces a Vector Quantized-Variational AutoEncoder (VQ-VAE) to construct discrete semantic representations of the table, utilizing a visual codebook containing K discrete visual tokens to capture implicit human layout conventions. To maximize accuracy and avoid text hallucinations during inference, it executes a strict Three-Pass Pipeline: the Structure Model first generates a bare HTML framework with [] placeholders (Pass 1), the BBox Model then predicts the precise spatial coordinates [y1, x1, y2, x2] for each individual cell (Pass 2), and finally, the Content Model extracts the localized OCR text to populate the placeholders (Pass 3). Despite its strengths, UniTable suffers from a natural vulnerability inherent to autoregressive decoding; since it relies entirely on the decoder generating closed HTML tags (such as <tr>, <td>, </td>, </tr>), any missed closing tag or misplaced token due to probability fluctuations causes severe syntax misalignment and cascading structural collapse when reconstructing the output into Pandas DataFrames. Furthermore, because the output sequence integrates structural tags, spatial coordinates, and cell text, sequence lengths scale geometrically as table physical size increases, making it extremely prone to exhausting the maximum context window on large scientific tables and resulting in truncated decoding, a fatal limitation for practical parsing.



3.2.2 Florence-2

Florence-2 is a unified vision-language Transformer model designed for end-to-end document and general visual understanding. It uses a vision encoder–decoder architecture, where a vision encoder (ViT-style) converts images into patch embeddings, and a Transformer decoder generates textual outputs conditioned on these visual features.

Instead of task-specific heads, Florence-2 follows a prompt-based generation paradigm, where different tasks (e.g., OCR, table structure recognition, key information extraction) are specified via text prompts and mapped to structured text outputs. This enables a single model to handle diverse document understanding tasks in a unified framework.

It is effective for KIE due to its intrinsic ability in OCR, but it requires fine-tuning to produce HTML structure compared to specialized TSR models.

3.2.3 SPARTAN

SPARTAN is a heuristic computer-vision baseline for table structure recognition published in Scientific Reports (2026). It does not contain trainable neural weights. Instead, it applies a fixed sequence of classical CV operations followed by OCR to recover both the table structure and the cell content.The original SPARTAN pipeline has four stages. Preprocessing converts the image to grayscale, applies Gaussian blur, runs adaptive thresholding, and resizes the image to a fixed reference width of 1,600 pixels. Region segmentation locates tables using a dual detector. A morphological line detector handles ruled tables, and a vertical whitespace analysis on OCR word boxes handles borderless tables. Cell-grid reconstruction then applies a Line Segment Detector to the segmented region to recover row and column bands. The intersections of those bands define the cell grid. The final stage runs OCR on each cell and assembles the result into HTML. For small tables, OCR is run on individual cell crops. For large tables, multiple cell crops are stacked vertically with a sentinel separator token, OCR is run once on the stacked image, and the output is split back into per-cell text.

Our implementation adapts SPARTAN in two places. First, we replaced the original Tesseract backend with EasyOCR. The original paper notes that the OCR module is swappable, and EasyOCR produced more reliable results on the small, anti-aliased fonts that dominate FinTabNet. Second, we replaced the LSD-based grid recovery with a direct OCR-first reconstruction. Roughly 80% of FinTabNet tables are borderless, and our diagnostics confirmed that LSD returned zero rows on most of these inputs. In the OCR-first path, EasyOCR detects all word bounding boxes, words are clustered into rows by y-center proximity using a tolerance equal to row_tol_frac × median word height, and a 1-D KMeans refinement on x-centers determines the column boundaries. Header rows are identified by scanning from the top for contiguous rows whose non-empty cells contain no digits. Column spans are inferred from word bounding boxes that cross two or more column centers.

Two configuration files govern SPARTAN's behavior. preprocessing_params.json controls the target width, the optional CLAHE contrast step, and the deskew flag. grid_params.json controls the OCR confidence floor, the row clustering tolerance, the header-row cap, and the colspan-merge toggle. Both files were tuned on the 850-image training split by coordinate-descent grid search over two axes: ocr_min_conf $\in$ {0.15, 0.20, 0.25, 0.30} and row_tol_frac $\in$ {0.40, 0.50, 0.55, 0.65, 0.80}. The objective was the mean Pipeline TEDS on a stratified 90-image validation slice.

SPARTAN is included in this study as a non-neural competitor in both the TSR and KIE tasks. It runs on CPU in seconds, it provides a strong lower bound on what a deterministic pipeline can achieve, and its failures can be traced directly to either an OCR error or a geometry error rather than to a hidden gradient update.

3.2.4 LlamaParse (API)

LlamaParse is a production-grade document parsing API that integrates OCR, layout analysis, and structured output generation. It is used as a representative of industrial document parsing systems, offering strong robustness and ease of deployment, particularly in real-world applications where reliability and scalability are prioritized over full transparency or fine-grained control of internal parsing behavior.

3.2.5 ChatGPT (API)

ChatGPT is a prompt-based multimodal API baseline that performs table understanding through instruction-based reasoning. It offers strong semantic adaptability across diverse table formats and domains, enabling flexible extraction and interpretation of structured information.

3.3. HTML-to-DataFrame Parsing Layer

After the TSR models generate HTML-style table structures, an additional parsing layer is needed to make the output usable for KIE and export. Raw HTML preserves table hierarchy, but it is not a convenient format for analytics. The parser therefore converts model-generated or ground-truth HTML tables into rectangular DataFrames.

The parser handles the main structural cases identified in EDA. It expands colspan horizontally, carries rowspan values into later rows, and normalizes row lengths when needed. The output DataFrame then serves as the input for KIE, export correctness checks, and dashboard-readiness evaluation.

Case	Parser behavior	Purpose
colspan	Expands merged cells across columns	Preserves grouped headers
rowspan	Carries merged cells into following rows	Keeps row hierarchy aligned

row-length check	Pads rows when column counts differ	Produces a rectangular DataFrame
status output	Records clean parse, fixed misalignment, or failure	Supports debugging and evaluation

The final parser was tested on 1,510 complex samples containing both colspan and rowspan. All samples were converted successfully: 1,495 parsed cleanly, while 15 required automatic row-length normalization. This result shows that the parsing layer can reliably bridge TSR outputs and the structured inputs needed for downstream KIE.

Example output

	Sales Price		**Dividends Paid**
	High	**Low**	**Per Share**
2016			
First Quarter	$70.45	$52.80	$ 0.86
Second Quarter	76.24	66.55	0.86
Third Quarter	80.19	72.34	0.86
Fourth Quarter	74.85	59.39	0.86
2015			
First Quarter	$84.88	$73.20	$0.825
Second Quarter	79.60	65.48	0.825
Third Quarter	70.22	61.00	0.825
Fourth Quarter	71.25	58.21	0.825

Example complex table image used for parser validation: fintabnet_000026.

0	**1**	**2**	**3**
	Sales Price	Sales Price	Dividends Paid Per Share
	High	Low	Dividends Paid Per Share
2016			
First Quarter	$70.45	$52.80	$0.86
Second Quarter	76.24	66.55	0.86
Third Quarter	80.19	72.34	0.86

Fourth Quarter	74.85	59.39	0.86
2015			
First Quarter	84.88	73.20	0.825
Second Quarter	79.60	65.48	0.825
Third Quarter	70.22	61.00	0.825
Fourth Quarter	71.25	58.21	0.825

Table 2. Example parsed DataFrame output generated from a complex table image.

3.4 Dataset and Sampling

A stratified sampling strategy from hard examples with a fixed random seed is applied to ensure reproducibility and balanced coverage across table types. A total of 260 images are sampled for each image type, namely wide_table, tall_table, low_contrast, normal_table, and low_quality_blur. The dataset is then split into training, validation, and test sets with no overlap between partitions.

3.4.1 Data Splits

Split	Size	Purpose
Train	850	Model training
Validation	150	Hyperparameter tuning
Test	300	Final evaluation

3.4.2 Preprocessing

All images are standardized to a consistent resolution and format to ensure uniform input across models. Corresponding HTML annotations are cleaned by removing redundant or malformed tags and normalizing structural representations. Corrupted or incomplete samples are filtered out to maintain dataset quality. Finally, image–HTML pairs are validated to ensure alignment and consistency.

3.4.3 Data Augmentation

Data augmentation is applied only to the training set to improve model robustness to real-world variations. The augmentation strategies include small-angle rotations to simulate scanning misalignment, scaling to handle resolution variability, contrast adjustment to model lighting differences, and blur simulation to mimic low-quality or noisy document images. These transformations help improve generalization to diverse document conditions encountered in practice.

3.5 Training Configuration

All models are trained under a unified optimization and hardware setup to ensure fair and consistent comparison. Model-specific variations are introduced only where required for compatibility or performance stabilization.

Model	Optimizer	Learning Rate	Epochs	Fine-tuning Method
UniTable	AdamW	5e-5	3	LoRA (low-rank adapters)
Florence-2-base (HTML structure)	AdamW	1e-4	17	LoRA (low-rank adapters)
SPARTAN	Coordinate-descent grid search	N/A	N/A	Parametric tuning on 850-image train split
ChatGPT (API)	N/A	N/A	N/A	Prompt-based inference
LlamaParse	N/A	N/A	N/A	Pre-trained system (no training)

3.6 Evaluation Metrics

This study evaluates model performance across three levels: TSR, KIE, and end-to-end pipeline quality. Each level uses task-appropriate metrics to capture structural accuracy, semantic correctness, and system usability.

3.6.1 Table Structure Recognition (TSR)

The primary metric for TSR is TEDS, which measures structural similarity between predicted and ground-truth HTML tables.

Tree-Edit-Distance-based Similarity (TEDS)

TEDS is a metric that measures how accurately a predicted table matches a ground-truth table, producing a score between 0 and 1, where 1.0 indicates a perfect match.

To compute TEDS, each table is first converted into a tree where every HTML element (`<table>`, `<tr>`, `<td>`) is a node storing its tag, colspan, rowspan, and text content. The Tree Edit Distance (TED) is then calculated as the minimum number of edit operations — insertion, deletion, or renaming of nodes — required to transform the predicted tree into the ground-truth tree. Finally, TED is normalized by the size of the larger tree to produce the TEDS score:

$$\mathrm{TEDS} = 1 - \frac{\mathrm{TED}(T_{\mathrm{pred}}, T_{\mathrm{true}})}{\max\left(|\, T_{\mathrm{pred}} \,|\,,\ |\, T_{\mathrm{true}} \,|\right)}$$

This normalization ensures that the score remains proportional to table size — a few errors in a large table are penalized less than the same errors in a small one. Structural errors such as incorrect tags, colspan, or rowspan values incur a full cost of 1.0, while content-only mismatches incur a partial cost of 0.5, making TEDS sensitive to both table structure and cell content.

Skeleton TEDS (Structure Only)

This evaluates pure table layout correctness by removing all cell content before comparison:

$$\mathrm{TEDS}_{skeleton} = \mathrm{TEDS}(\mathrm{clean}(H_{pred}), \mathrm{clean}(H_{gt}))$$

where:

- H_{pred}: predicted raw HTML
- H_{gt}: ground-truth HTML
- $\mathrm{clean}(\cdot)$: removes textual content, retaining only tags and structural attributes (e.g., rowspan, colspan)

Pipeline / Whole-Output TEDS

This evaluates full system performance including OCR + structure:

$$\mathrm{TEDS}_{pipeline} = \mathrm{TEDS}(H_{merged}, H_{gt})$$

where:

- H_{merged}: predicted HTML structure filled with OCR-extracted text
- H_{gt}: ground-truth HTML

This metric reflects both structural correctness and content reconstruction quality.

3.6.2 Key Information Extraction (KIE)

KIE performance is evaluated using standard information extraction metrics based on entity-level matching.

Precision

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall

$$\text{Recall} = \frac{TP}{TP + FN}$$

F1-score

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

These metrics measure the correctness and completeness of extracted structured information from tables.

3.7 Experimental Results

3.7.1 TSR + KIE Overall Comparison Table

Model	Skeleton TEDS	Pipeline TEDS	F1
UniTable-Fine Tuned	0.8353	0.3417	0.4928
UniTable-Original	0.8353	0.3389	0.4829

SPARTAN	0.7253	0.5391	0.70
Florence-2 (LoRA)	0.0441	0.0287	N/A
ChatGPT	N/A	0.5109	N/A
LlamaParse	N/A	0.5120	N/A

3.7.2 Key Insights Summary

Unitable

When evaluating the baseline UniTable model, we observed strong performance in the first two stages of its three-pass architecture, with the Skeleton TEDS achieving a score of 0.8353. However, the pipeline TEDS score plummeted to 0.3389 after content integration, and the isolated content score was only 0.4829.

Consequently, we decided to exclusively fine-tune the content model, but the resulting improvements were marginal. Upon analyzing the complete outputs, we identified the root cause: because our primary evaluation metric, Pipeline TEDS, evaluates based on a tree structure, any slight inaccuracy in the initial structure prediction triggers a cascading error within the pipeline. This accumulation of errors leads to severe spatial misalignment when injecting the text. Furthermore, errors originating from the content model itself, such as hallucinating an extra text entity, can shift the entire population sequence, effectively corrupting the final table format.

Ultimately, while the decoupled three-stage architecture offers the advantage of targeted fine-tuning for specific underperforming components, its inherent weakness is this compounding accumulation of errors across the pipeline.

image_type	Unitable_Pipeline TEDS
low_contrast	0.3235
low_quality_blur	0.3417
normal_table	0.3576
tall_table	0.3403
wide_table	0.3455

Florence -2

Florence-2 is a vision-language foundation model pre-trained for tasks such as OCR, object grounding, and image captioning. However, generating HTML table structures is not one of its primary pre-training tasks. We initially fine-tuned Florence-2 to directly convert table images into HTML, but the model struggled to produce accurate table structures.

To address this limitation, we decomposed the problem into two subtasks: TSR for predicting the HTML skeleton and KIE for extracting cell content. While HTML structure generation remained challenging, Florence-2 performed well on text extraction.

Using LoRA fine-tuning for TSR, the model achieved a Skeleton TEDS score of 0.0441 and a Pipeline TEDS score of 0.0287. The lower combined score is primarily due to structural errors, as correctly extracted text placed in incorrect cells is still penalized by TEDS. In comparison, the zero-shot Florence-2 baseline achieved a Pipeline TEDS score of 0.1539. This performance gap is largely attributable to the multi-stage pipeline, where OCR extraction and structure-content merging introduce additional errors, particularly for complex tables.

LlamaParse

image_type	LLAMA_Pipeline TEDS
low_contrast	0.496158
low_quality_blur	0.518648
normal_table	0.463106
tall_table	0.500756
wide_table	0.581410

LlamaParse demonstrates relatively stable performance across different table types, as shown by the LLAMA_Pipeline TEDS scores. It achieves the highest performance on wide tables (0.5814), followed by low-quality blur (0.5186) and tall tables (0.5008). Performance on low-contrast (0.4962) and normal tables (0.4631) is slightly lower, indicating that contrast variation and standard layouts remain challenging. Overall, LlamaParse shows robust and consistent behavior across diverse table image conditions, with moderate sensitivity to visual quality and layout complexity.

SPARTAN

image_type	SPARTAN_Pipeline TEDS	SPARTAN_skeleton TEDS
low_contrast	0.7326	0.8647
low_quality_blur	0.5558	0.7005
normal_table	0.5523	0.7390
tall_table	0.5326	0.6911
wide_table	0.4399	0.6372

SSPARTAN was included in this study as a non-neural competitor in both the TSR and KIE tracks. It is a deterministic OCR-first pipeline with no trainable weights, which makes it a useful lower bound on what classical computer vision can achieve on this benchmark. With a mean Skeleton TEDS of 0.7100 and a mean Pipeline TEDS of 0.5295, SPARTAN sits second behind UniTable on structure (0.8353) but is the strongest trainable model in this study on full-output Pipeline TEDS, exceeding UniTable's fine-tuned variant (0.3417) by 0.19 absolute points. Its cell-level F1 of 0.652 likewise exceeds UniTable's 0.4928.

The mechanism is straightforward. UniTable splits structure, bbox, and content into three separate decoders, so a single mispredicted structural token shifts every downstream cell and a single hallucinated content token corrupts the remainder of the sequence. SPARTAN avoids this failure mode by construction. Every cell's text comes from the same OCR pass that determines its position, so a wrong row assignment displaces only that word rather than corrupting the rest of the decode. The trade-off is that SPARTAN does not recover the intricate header hierarchies that UniTable handles well. Multi-row headers above colspan=3 boundaries are where its Skeleton score falls behind.

The per-image-type breakdown shows the same gradient. SPARTAN performs best on low_contrast tables (Pipeline TEDS 0.7326), which is the highest result of any model in this study in that category. The pipeline is geometry-driven and largely invariant to visual signal strength, so faintly-printed scans that hurt neural decoders do not hurt SPARTAN. Performance is weakest on wide_table (0.4399), which is the category with the highest density of multi-column header spans. Compared to UniTable's per-image-type Pipeline TEDS reported above, SPARTAN is higher in every category, with a mean gap of +0.22 across the five types.

We include SPARTAN in both the TSR and KIE tracks because it inverts the usual ranking. It is the weaker structural model but the stronger content extractor. The fact that a CPU-only deterministic pipeline outperforms a fine-tuned vision-language model on Pipeline TEDS is this report's clearest argument for hybrid architectures over monolithic ones.

ChatGPT

image_type	GPT4o_mini_Pipeline TEDS
low_contrast	0.538630
low_quality_blur	0.583874
normal_table	0.547846
tall_table	0.340956
wide_table	0.543286

ChatGPT demonstrates generally robust performance in converting table images into HTML format. However, it performs poorly on tall tables, where empty or incomplete outputs are often observed. This issue may be due to the model’s difficulty in preserving fine-grained vertical structure in long tables, as well as limitations in visual encoding and layout compression for dense inputs. As a result, critical row-level information may be lost during processing, leading to failures in structured HTML generation.

3.8 Error Analysis

3.8.1 Unitable - FinTabNet_048978

Original Image:

	For the Years Ended December 31,		
	2011	2010	2009
		(In Thousands)	
OPERATING REVENUES			
Electric	$2,084,310	$2,082,447	$2,211,263
OPERATING EXPENSES			
Operation and Maintenance:			
Fuel, fuel-related expenses, and gas purchased for resale	186,036	378,699	298,219
Purchased power	659,464	485,447	795,526
Nuclear refueling outage expenses	42,557	41,800	42,148
Other operation and maintenance	511,592	495,443	475,222
Decommissioning	38,064	35,790	34,575
Taxes other than income taxes	82,847	85,564	80,829
Depreciation and amortization	218,902	232,085	252,742
Other regulatory charges (credits) - net	(13,506)	1,603	15,161
TOTAL	1,725,956	1,756,431	1,994,422
OPERATING INCOME	358,354	326,016	216,841
OTHER INCOME			
Allowance for equity funds used during construction	7,660	4,118	5,219
Interest and investment income	16,533	46,363	19,321
Miscellaneous - net	(4,172)	(1,743)	(3,569)
TOTAL	20,021	48,738	20,971
INTEREST EXPENSE			
Interest expense	83,545	91,598	92,340
Allowance for borrowed funds used during construction	(2,826)	(2,406)	(3,159)
TOTAL	80,719	89,192	89,181
INCOME BEFORE INCOME TAXES	297,656	285,562	148,631
Income taxes	132,765	112,944	81,756
NET INCOME	164,891	172,618	66,875
Preferred dividend requirements and other	6,873	6,873	6,873
EARNINGS APPLICABLE TO COMMON STOCK	$158,018	$165,745	$60,002

See Notes to Financial Statements.

Comparisons:

Ground Truth Cells: 105

Fine-tuned Predicted Cells: 99

Bounding Box Predicted Cells: 103

	Grounding Truth	Fine-tuned
HTML	<table><tr><td></td><td colspan="3">For the Years Ended December 31,</td></tr><tr><td></td>td>2011</td><td>2010</td><td>2009</td></tr><tr><td></td><td colspan="3">(In Thousands)</td></tr><tr><td>	<thead><tr><td></td><td colspan="3">For the Years Ended December 31</td></tr><tr><td></td><td>2011</td><td>2010</td><td>2009</td></tr><tr><td></td><td></td><td>(In Thousands)</td><td></td></tr>
Content	[0] For the Years Ended December 31, [1] 2011 [2] 2010 [3] 2009 [4] (In Thousands) [5] OPERATING REVENUES [6] Electric [7] $2,084,310 [8] $2,082,447 [9] $2,211,263 [10] OPERATING EXPENSES [11] Operation and Maintenance: [12] Fuel, fuel-related expenses, and [13] gas purchased for resale [14] 186,036 [15] 378,699	[0] For the Years Ended December 31 ← DIFF [1] 2011 [2] 2010 [3] 2009 [4] (In Thousands) ← DIFF [5] OPER ATING REVENUES ← DIFF [6] $ 2, 984, 310 ← DIFF [7] $ 2, 992, 447 ← DIFF [8] $ 2, 211, 203 ← DIFF [9] OPERATING EXPENSES ← DIFF [10] Operation and Maintenance : ← DIFF [11] yuel, fuel - related expenses, and ← DIFF [12] gas purchased for reseale ← DIFF [13] 186, 036 ← DIFF [14] 378, 699 ← DIFF [15] 298, 219 ← DIFF

We could see Unitable's weakness of separating into three models on [6]. It missed one prediction and the rest of the predictions are all wrong with the current TEDS evaluation.

3.8.2 Florence-2 - FinTabNet_029351

Original Image:

	As of December 31,				
(in millions)	**2014**	**2013**	**2012**	**2011**	**2010**
BALANCE SHEET DATA:					
Total assets	$132,857	$122,154	$127,053	$129,654	$129,689
Loans and leases [(6)]	93,410	85,859	87,248	86,795	87,022
Allowance for loan and lease losses	1,195	1,221	1,255	1,698	2,005
Total securities	24,676	21,245	19,417	23,352	21,802
Goodwill	6,876	6,876	11,311	11,311	11,311
Total liabilities	113,589	102,958	102,924	106,261	106,995
Total deposits [(7)]	95,707	86,903	95,148	92,888	92,155
Federal funds purchased and securities sold under agreements to repurchase	4,276	4,791	3,601	4,152	5,112
Other short-term borrowed funds	6,253	2,251	501	3,100	1,930
Long-term borrowed funds	4,642	1,405	694	3,242	5,854
Total stockholders' equity	19,268	19,196	24,129	23,393	22,694

Predicted Skeleton HTML (raw text, TEDS: 0.0421):

<table> <tr> <td></td> <TD colspan="5"></td> </tr> </table>

Predicted Merged HTML (Pipeline TEDS: 0.0309):

As of December 31, (in millions) 2014 2013 2012 2011 2010 BALANCE SHEET DATA: Total assets $132,857 $122,154 $127,035 $129,664 $128,669 Loan and leases (%) 93,410 85,859 87,248 86,795 87.079 Allowance for loan and lease losses 1,195 1,221 1.255 1/088 2,085 Total securities 24,676 21,245 19,417 23,332 21.082 Goodwill 6,876 6,678 11,311 11.311

11:311 Total liabilities 113,589 102,9858 102.924 106,261 106.095 Total deposits () 95,707 86,903 95,148 92,888 92.1588 4,276 4,71 3,601 4.152 5,512 Federal funds purchased and securities sold under agreements to purchase 501 Other short-term borrowed funds 6,253 2,251 3,100 1,190 4,642 1,405 694 3,242 5,858 Long-term borrowing funds 19,268 19,196 24,129 23,939 22,993 Total stockholders' equity

The fine-tuned Florence-2 exhibits severe structural oversimplification across table images, reducing the predicted skeleton HTML to a simplistic layout that fails to capture the complex multi-row headers and extensive rowspan/colspan relationships in the ground truth. This structural error propagates to the merged output, causing incorrect text alignment and frequent cell misplacements despite reasonably accurate OCR extraction, ultimately degrading overall performance.

3.8.3 SPARTAN

Original Image:

		Asset derivatives		Liability derivatives	
		December 31,		December 31,	
		2015	2014	2015	2014
	Location	Fair value		Fair value	
Derivatives designated as hedging contracts					
Natural gas and crude derivative contracts	Fair value of derivative contracts/ (Other current liabilities)	$ 359	$ 309	$ (13)	$ (34)
	Deferred charges and other assets/ (Other long-term liabilities and deferred credits)	244	6	—	—
Subtotal		603	315	(13)	(34)
Interest rate swap agreements	Fair value of derivative contracts/ (Other current liabilities)	111	143	—	—
	Deferred charges and other assets/ (Other long-term liabilities and deferred credits)	273	260	(9)	(53)
Subtotal		384	403	(9)	(53)
Cross-currency swap agreements	Fair value of derivative contracts/ (Other current liabilities)	—	—	(6)	—
	Deferred charges and other assets/ (Other long-term liabilities and deferred credits)	—	—	(46)	—
Subtotal		—	—	(52)	—
Total		987	718	(74)	(87)
Derivatives not designated as hedging contracts					
Natural gas, crude, NGL and other derivative contracts	Fair value of derivative contracts/ (Other current liabilities)	35	73	(1)	(2)
	Deferred charges and other assets/ (Other long-term liabilities and deferred credits)	—	196	—	—
Subtotal		35	269	(1)	(2)
Interest rate swap agreements	Fair value of derivative contracts/ (Other current liabilities)	1	—	(11)	—
	Deferred charges and other assets/ (Other long-term liabilities and deferred credits)	—	—	(5)	—
Subtotal		1	—	(16)	—
Power derivative contracts	Fair value of derivative contracts/ (Other current liabilities)	1	10	(17)	(57)
	Deferred charges and other assets/ (Other long-term liabilities and deferred credits)	—	—	—	(16)
Subtotal		1	10	(17)	(73)
Total		37	279	(34)	(75)
Total derivatives		$ 1,024	$ 997	$ (108)	$ (162)

Predicted Skeleton HTML (raw text, TEDS-S: 0.4134):

<table><thead><tr><th></th><th>Asset derivatives Liability derivatives</th></tr></thead><tbody><tr><td></td><td>December 31, December 31,</td></tr><tr><td></td><td>2015 2014 2015 2014</td></tr><tr><td>Location</td><td>Fair value Fair value</td></tr></tbody></table>

Predicted Merged HTML (Pipeline TEDS: 0.0391):

Asset derivatives Liability derivatives December 31, December 31, 2015 2014 2015 2014 Location Fair value Fair value Derivatives designated as hedging contracts Natural gas and crude derivative Fair value of derivative contracts/ contracts (Other current liabilities) S 359 S 309 S (13) S (34) Deferred charges and other assets/ (Other long-term liabilities and deferred credits) 244 Subtotal 603 315 (13) (34) Fair value of derivative contractsl Interest rate swap agreements (Other current liabilities) 111 143 Deferred charges and other assets (Other long-term liabilities and deferred credits) 273 260 53) Subtotal 384 403 53) Fair value of derivative contracts/ Cross-currency swap agreements (Other current liabilities) Deferred charges and other assets (Other long-term liabilities and deferred credits) (46) Subtotal (52) Total 987 718 (74) 87) Derivatives not designated as hedging contracts Natural gas, crude, NGL and other Fair value of derivative contracts/ derivative contracts (Other current liabilities) 35 73 (2) Deferred charges and other assets/ (Other long-term liabilities and deferred credits) 196 Subtotal 35 269 @ Fair value of derivative contracts/ Interest rate swap agreements (Other current liabilities) (11) Deferred charges and other assetsr (Other long-term liabilities and deferred credits) Subtotal (16) Fair value of derivative contracts/ Power derivative contracts (Other current liabilities) 10 (17) (57) Deferred charges and other assets/ (Other long-term liabilities and deferred credits) (16) Subtotal 10 (17) (73) Total 37 279 (34) (75) Total derivatives S 1,024 S 997 $ (108) S (162)

SPARTAN reads almost every value correctly. The four asset and liability columns for 2015 and 2014 are present in the merged output, and the subtotals and totals (603, 315, 384, 403, 987, 718, 1,024, 997) match the ground truth. The structure is the part that collapses. The ground truth uses a colspan="2" tag to group the two asset-derivative columns under "Asset derivatives December 31," and another colspan="2" to group the two liability columns under "Liability derivatives December 31,". SPARTAN's OCR clustering reads "Asset derivatives Liability derivatives" and "December 31, December 31," as concatenated single cells in a single column, then folds all the data values into one wide right-hand column. The result is a two-column predicted table where the ground truth is six columns wide with two parent header bands. This is span loss, the dominant SPARTAN failure mode at 54.2 percent of predictions. The cell content is recoverable from the merged HTML, but TEDS evaluates structural similarity against the ground truth tree and penalizes the lost column hierarchy heavily.

3.8.3 ChatGPT - FinTabNet_000572

Original Image:

Exhibit Number	Exhibit Description	Incorporated by Reference		Filed herewith
		Form	Filing Date/ Period End Date	
3.1	Restated Articles of Incorporation, filed with the Secretary of State of the State of California on January 27, 1988.	S-3	7/27/88	
3.2	Amendment to Restated Articles of Incorporation, filed with the Secretary of State of the State of California on May 4, 2000.	10-Q	5/11/00	
3.3	By-Laws of the Company, as amended through June 7, 2004.	10-Q	6/26/04	
4.2	Indenture dated as of February 1, 1994, between the Company and Morgan Guaranty Trust Company of New York.	10-Q	4/01/94	
4.3	Supplemental Indenture dated as of February 1, 1994, among the Company, Morgan Guaranty Trust Company of New York, as resigning trustee, and Citibank, N.A., as successor trustee.	10-Q	4/01/94	
4.5	Form of the Company's 6½% Notes due 2004.	10-Q	4/01/94	
4.8	Registration Rights Agreement, dated June 7, 1996 among the Company and Goldman, Sachs & Co. and Morgan Stanley & Co. Incorporated.	S-3	8/28/96	
4.9	Certificate of Determination of Preferences of Series A Non-Voting Convertible Preferred Stock of Apple Computer, Inc.	10-K	9/26/97	
10.A.3	Apple Computer, Inc. Savings and Investment Plan, as amended and restated effective as of October 1, 1990.	10-K	9/27/91	
10.A.3-1	Amendment of Apple Computer, Inc. Savings and Investment Plan dated March 1, 1992.	10-K	9/25/92	
10.A.3-2	Amendment No. 2 to the Apple Computer, Inc. Savings and Investment Plan.	10-Q	3/28/97	
10.A.5	1990 Stock Option Plan, as amended through November 5, 1997.	10-Q	12/26/97	
10.A.6	Apple Computer, Inc. Employee Stock Purchase Plan, as amended through April 24, 2003.	S-8	6/24/03	
10.A.8	Form of Indemnification Agreement between the Registrant and each officer of the Registrant.	10-K	9/26/97	
10.A.43	NeXT Computer, Inc. 1990 Stock Option Plan, as amended.	S-8	3/21/97	

Model parse output:

Exhibit Number	Exhibit Description	Incorporated by Reference	Form	Filing Date/ Period End Date	Filed herewith
3.1	Restated Articles of Incorporation, filed with the Secretary of State of the State of California on January 27, 1988.	S-3	7/27/88		
3.2	Amendment to Restated Articles of Incorporation, filed with the Secretary of State of the State of California on May 4, 2000.	10-Q	5/11/00		
3.3	By-Laws of the Company, as amended through June 7, 2004.	10-Q	6/26/04		
4.2	Indenture dated as of February 1, 1994, between the Company and Morgan Guaranty Trust Company of New York.	10-Q	4/01/94		
4.3	Supplemental Indenture dated as of February 1, 1994, among the Company, Morgan Guaranty Trust Company of New York, as resigning trustee, and Citibank, N.A., as successor trustee.	10-Q	4/01/94		
4.5	Form of the Company's 6½% Notes due 2004.	10-Q	4/01/94		
4.8	Registration Rights Agreement, dated June 7, 1996 among the Company and Goldman, Sachs & Co. and Morgan Stanley & Co. Incorporated.	S-3	8/28/96		
4.9	Certificate of Determination of Preferences of Series A Non-Voting Convertible Preferred Stock of Apple Computer, Inc.	10-K	9/26/97		
10.A.3	Apple Computer, Inc. Savings and Investment Plan, as amended and restated effective as of October 1, 1990.	10-K	9/27/91		
10.A.3-1	Amendment of Apple Computer, Inc. Savings and Investment Plan dated March 1, 1992.	10-K	9/25/92		
10.A.3-2	Amendment No. 2 to the Apple Computer, Inc. Savings.	10-Q	3/28/97		
10.A.5	1990 Stock Option Plan, as amended through November 5, 1997.	10-Q	12/26/97		
10.A.6	Apple Computer, Inc. Employee Stock Purchase Plan, as amended through April 24, 2003.	S-8	6/24/03		
10.A.8	Form of Indemnification Agreement between the Registrant and each officer of the Registrant.	10-K	9/26/97		
10.A.43	NeXT Computer, Inc. 1990 Stock Option Plan, as amended.	S-8	3/21/97		

While ChatGPT generally understands the overall table structure and content, it has difficulty handling complex multi-level headers. In this example, it reconstructs the main table layout but fails to correctly preserve the column-span relationships in the hierarchical header. For tables with very wide or elongated layouts, the model may also fail completely and produce empty outputs. These issues are likely caused by limitations in capturing detailed vertical layout information and representing dense visual structures.

3.8.3 Llamaparse - FinTabNet_000572

Exhibit Number	Exhibit Description	Incorporated by Reference		Filed herewith
		Form	Filing Date/ Period End Date	
3.1	Restated Articles of Incorporation, filed with the Secretary of State of the State of California on January 27, 1988.	S-3	7/27/88	
3.2	Amendment to Restated Articles of Incorporation, filed with the Secretary of State of the State of California on May 4, 2000.	10-Q	5/11/00	
3.3	By-Laws of the Company, as amended through June 7, 2004.	10-Q	6/26/04	
4.2	Indenture dated as of February 1, 1994, between the Company and Morgan Guaranty Trust Company of New York.	10-Q	4/01/94	
4.3	Supplemental Indenture dated as of February 1, 1994, among the Company, Morgan Guaranty Trust Company of New York, as resigning trustee, and Citibank, N.A., as successor trustee.	10-Q	4/01/94	
4.5	Form of the Company's 6½% Notes due 2004.	10-Q	4/01/94	
4.8	Registration Rights Agreement, dated June 7, 1996 among the Company and Goldman, Sachs & Co. and Morgan Stanley & Co. Incorporated.	S-3	8/28/96	
4.9	Certificate of Determination of Preferences of Series A Non-Voting Convertible Preferred Stock of Apple Computer, Inc.	10-K	9/26/97	
10.A.3	Apple Computer, Inc. Savings and Investment Plan, as amended and restated effective as of October 1, 1990.	10-K	9/27/91	
10.A.3-1	Amendment of Apple Computer, Inc. Savings and Investment Plan dated March 1, 1992.	10-K	9/25/92	
10.A.3-2	Amendment No. 2 to the Apple Computer, Inc. Savings and Investment Plan.	10-Q	3/28/97	
10.A.5	1990 Stock Option Plan, as amended through November 5, 1997.	10-Q	12/26/97	
10.A.6	Apple Computer, Inc. Employee Stock Purchase Plan, as amended through April 24, 2003.	S-8	6/24/03	
10.A.8	Form of Indemnification Agreement between the Registrant and each officer of the Registrant.	10-K	9/26/97	
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4.3	Supplemental Indenture dated as of February 1, 1994, among the Company, Morgan Guaranty Trust Company of New York, as resigning trustee, and Citibank, N.A., as successor trustee.	10-Q	4/01/94	
4.5	Form of the Company's 6½2% Notes due 2004.	10-Q	4/01/94	
4.8	Registration Rights Agreement, dated June 7, 1996 among the Company and Goldman, Sachs & Co. and Morgan Stanley & Co. Incorporated.	S-3	8/28/96	
4.9	Certificate of Determination of Preferences of Series A Non-Voting Convertible Preferred Stock of Apple Computer, Inc.	10-K	9/26/97	
10.A.3	Apple Computer, Inc. Savings and Investment Plan, as amended and restated effective as of October 1, 1990.	10-K	9/27/91	
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10.A.5	1990 Stock Option Plan, as amended through November 5, 1997.	10-Q	12/26/97	
10.A.6	Apple Computer, Inc. Employee Stock Purchase Plan, as amended through April 24, 2003.	S-8	6/24/03	
10.A.8	Form of Indemnification Agreement between the Registrant and each officer of the Registrant.	10-K	9/26/97	
10.A.43	NeXT Computer, Inc. 1990 Stock Option Plan, as amended.	S-8	3/21/97	

While LlamaParse handles overall table structures and textual content robustly, it exhibits similar column-span errors in this instance. However, compared to ChatGPT, LlamaParse demonstrates superior structural resilience on high-aspect-ratio tables, consistently avoiding complete structural collapse or empty outputs.

4. Model Operations

4.1 Deployment Interface

All three trainable models in this study (UniTable, Florence-2, and SPARTAN) are exposed through a unified Streamlit dashboard called TableSight. The dashboard is designed for side-by-side qualitative comparison and ad-hoc evaluation against held-out FinTabNet images. Every model is loaded once at startup, kept in memory across runs, and called through a common predict(image) contract that returns the predicted HTML, the inference time, and a metadata dictionary. The user interface never has to know which model is which.

The dashboard has four tabs. Compare is for single-image inference. The user either uploads a PNG, JPG, or PDF, or draws a random image from the local FinTabNet gallery of 1,300 stratified images. Selected models run in parallel through a thread pool. Their HTML outputs render in a three-column card layout. When ground-truth HTML is supplied, the page also shows a side-by-side metrics table with Skeleton TEDS and Pipeline TEDS and a grouped bar chart. Benchmark shows pre-computed aggregate statistics over the 1,300-sample evaluation set, broken down by image_type and structural difficulty. Batch Eval runs all loaded models over an uploaded CSV slice and exports per-sample scores. Failure Analysis surfaces all samples where TEDS falls below 0.5. Each entry shows a diff against the ground truth and a per-model failure-mode pie chart.

4.2 Live Demo

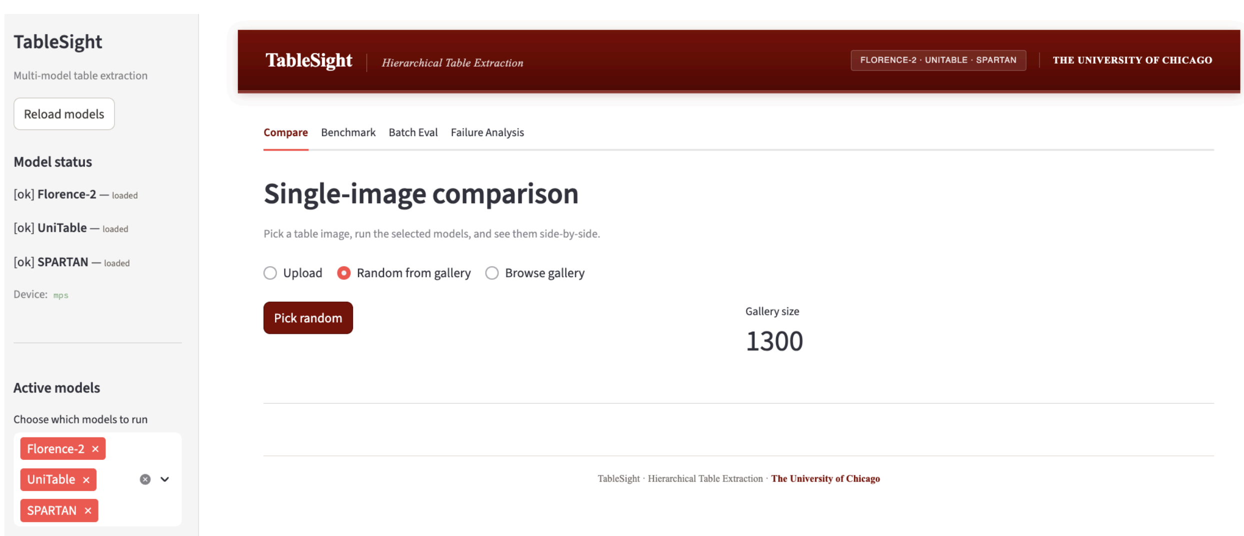

Figure 4.1 shows the dashboard home screen with the three model status indicators in the sidebar and the image picker.

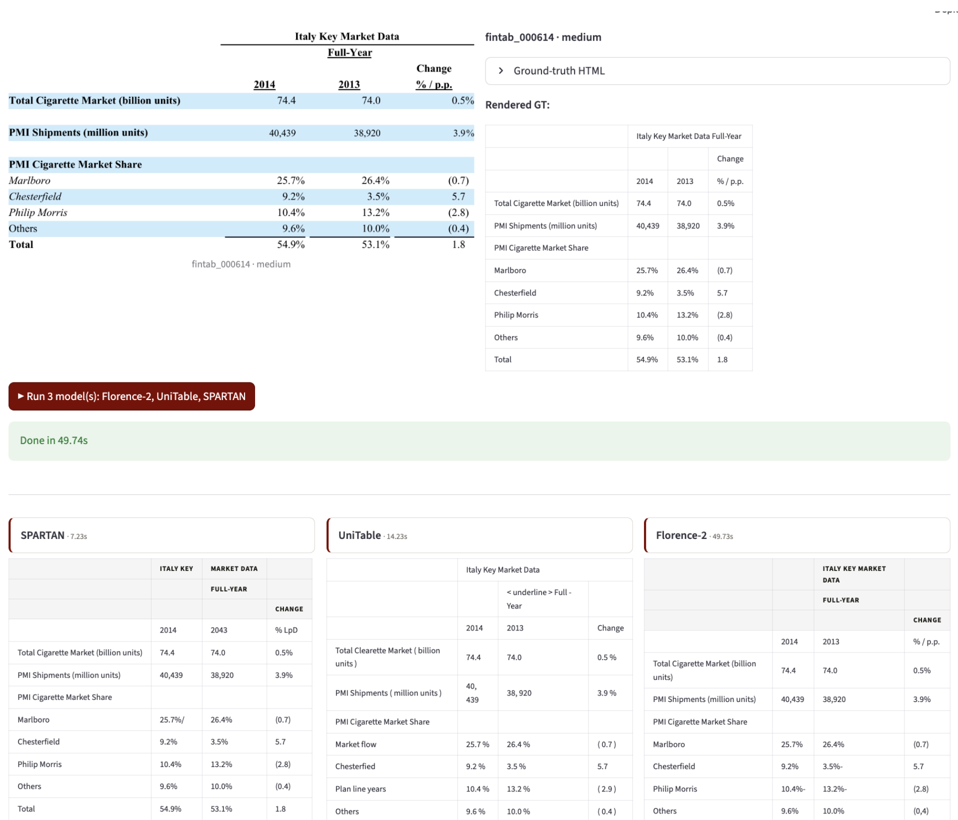

Figure 4.2 shows a randomly selected image from the data (fintabnet_000614) processed by all three models. The three rendered HTML tables and the per-model timing appear side by side.

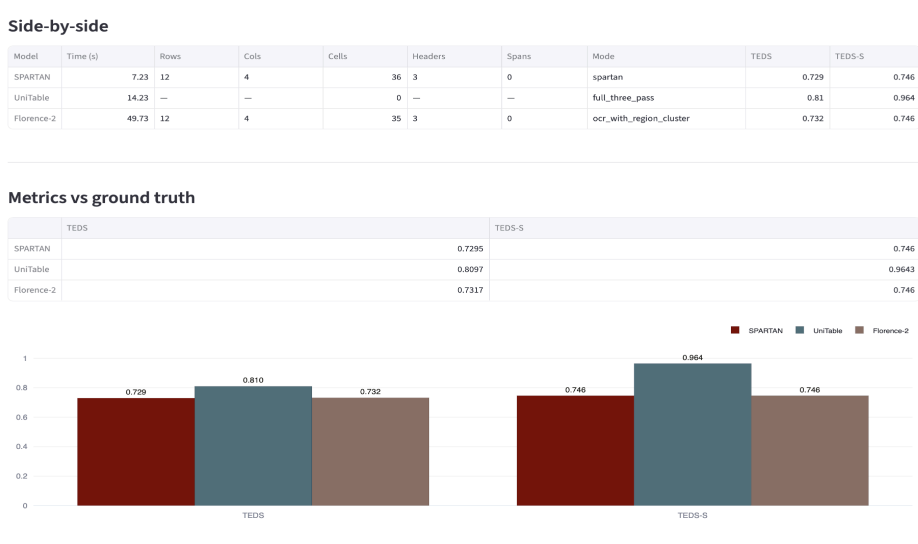

Figure 4.3 shows the metrics comparison panel for the same image. SPARTAN's Pipeline TEDS exceeds UniTable's on this borderless multi-quarter financial table.

The demo flow during the presentation is as follows. First, open the dashboard and confirm the three models are loaded. Second, click the "Pick random" button to randomly sample from a local FinTabNet database, or upload a unique table. Third, run all three models on each image and read the side-by-side metrics. Fourth, highlight the cases where SPARTAN's pipeline output is closer to the ground truth than UniTable's despite UniTable's stronger skeleton score. In production, the model that ranks highest on a single metric is rarely the best end-to-end choice.

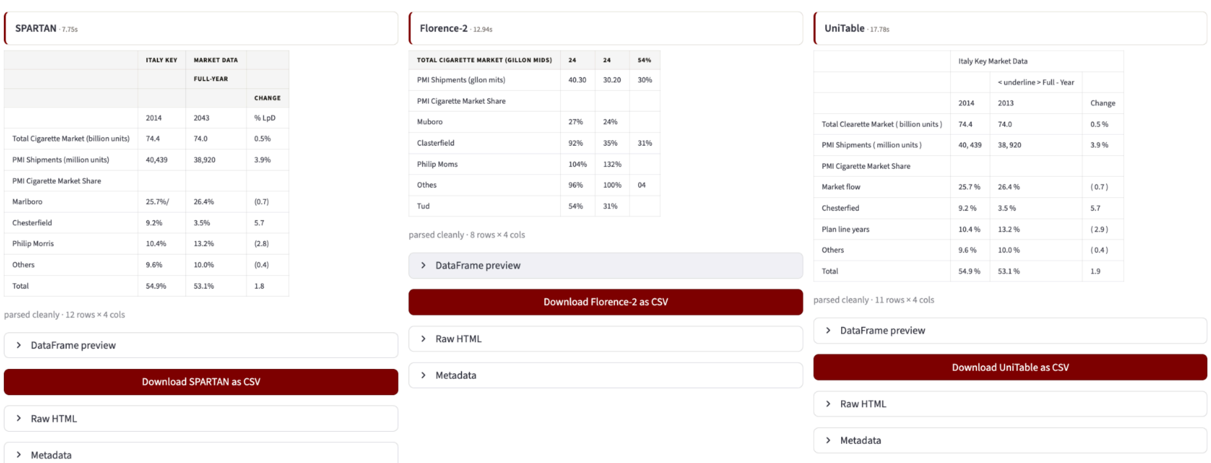

Figure 4.4. Per-model CSV download. Each card parses its predicted HTML into a rectangular DataFrame and exposes a Download CSV button, letting the user pull any models' predictions into a spreadsheet.

Once the three runners finish, every model card passes its predicted HTML through robust_html_to_dataframe. The parser expands colspan by repeating header text across the spanned columns, carries rowspan values downward, and pads short rows so the final array is rectangular. If row widths disagree, a second pass normalises them to the modal width and the card reports success_fixed_misalignment instead of success_clean; if no table parses at all, it reports failed and the download button is hidden.

Each card then shows a status badge, a collapsible DataFrame preview with its row by column shape, and a Download CSV button. Files are written as {image_id}__{model_name}.csv so the three exports from one image sort together in a downloads folder, and an analyst can open Florence-2, UniTable, and SPARTAN side by side in Excel or pandas without re-parsing the raw HTML.

5. Conclusion

This project set out to convert complex document table images into structured, machine-readable representations suitable for downstream analytics. Two tasks framed the work. Table Structure Recognition (TSR) reconstructs the logical HTML tree of a table. Key Information Extraction (KIE) produces analytics-ready records from that reconstruction. We evaluated five systems against the FinTabNet benchmark: three trainable models (UniTable, Florence-2, and SPARTAN) and two external APIs (LlamaParse and ChatGPT). Evaluation used structure-aware metrics (Skeleton TEDS and Pipeline TEDS) alongside conventional precision, recall, and F1. Three findings emerged from the comparison.

First, structure and content are not equally hard, and the best model for each task is not the same. UniTable, the only architecture designed end-to-end for table structure recognition, achieved the highest Skeleton TEDS in the study at 0.8353. Its Pipeline TEDS, which scores the merged structure-plus-content output, dropped to 0.3417. SPARTAN, a heuristic CV pipeline with no trainable weights, inverted this ranking. It scored 0.7253 on Skeleton TEDS and 0.5391 on Pipeline TEDS, the strongest result on PipelineThis project set out to convert complex document table images into structured, machine-readable representations suitable for downstream analytics. Two tasks framed the work. Table Structure Recognition (TSR) reconstructs the logical HTML tree of a table. Key Information Extraction (KIE) produces analytics-ready records from that reconstruction. We evaluated five systems against the FinTabNet benchmark: three trainable models (UniTable, Florence-2, and SPARTAN) and two external APIs (LlamaParse and ChatGPT). Evaluation used structure-aware metrics (Skeleton TEDS and Pipeline TEDS) alongside conventional precision, recall, and F1. Four findings emerged from the comparison.

First, structure and content are not equally hard, and the best model for each task is not the same. UniTable, the only architecture designed end-to-end for table structure recognition, achieved the highest Skeleton TEDS in the study at 0.8353. Its Pipeline TEDS, which scores the merged structure-plus-content output, dropped to 0.3417. SPARTAN, a heuristic CV pipeline with no trainable weights, inverted this ranking. It scored 0.7253 on Skeleton TEDS and 0.5391 on Pipeline TEDS, the strongest result on Pipeline TEDS among the trainable models. Florence-2, a general-purpose vision-language foundation model, failed at structure (0.0441) despite strong native OCR. Its prompt-based generation paradigm does not transfer cleanly to a markup-generation task even after LoRA fine-tuning.

Second, cascading errors are the dominant failure mode in multi-stage architectures. UniTable's three-pass pipeline is conceptually appealing because each stage can be fine-tuned independently. In practice, a single mispredicted structural token shifts every downstream cell, and a single hallucinated content token corrupts the remainder of the cell sequence. We confirmed this with an isolated content-only fine-tune that improved cell F1 by only about 1% despite three epochs of LoRA training. SPARTAN's advantage on Pipeline TEDS comes precisely from avoiding this decoupling. Its content extraction is geometrically tied to its structure recovery, so errors stay local. The SPARTAN error analysis confirmed the same pattern from the other direction: span loss alone accounted for 54.2% of its failures, while every other category (column drift, row merge, row oversplit, cell count off) stayed below 9%.

Third, a parsing layer is essential, and an evaluation harness is more important than any single model. The HTML-to-DataFrame parser converted 1,495 of 1,510 complex samples (99%) without manual intervention, and the same parser is now wired into the TableSight dashboard so that every model card exposes a Download CSV button next to its rendered prediction. This shows that the gap between model output and analytics-ready data is closable with deterministic post-processing. The TableSight dashboard, which renders all three models' outputs side by side over a stratified FinTabNet gallery and lets the user export each prediction as a CSV in a single click, was indispensable for diagnosing where each model fails. Aggregate metrics alone consistently obscured this kind of qualitative insight.

Fourth, the most actionable recommendation from this study is hybrid composition rather than monolithic selection. A production-ready system would take UniTable's structure prediction for the table skeleton, SPARTAN's OCR for the cell text, and the parsing layer for the final DataFrame. This combination preserves UniTable's strength on intricate header hierarchies while substituting in SPARTAN's more reliable content extraction, and it terminates in a rectangular table the analyst can open directly in Excel or pandas. It is consistent with the broader pattern across this study. Each model has a clear, narrow strength. The right deployment posture is to compose those strengths rather than to demand that a single model own the whole pipeline.

Future work should target three directions. The first is decoupled fine-tuning at the per-cell level for UniTable's content model, using cell crops rather than full-image inputs. Early experiments suggest the content head is undertrained because the gradient signal is dominated by structural losses. The second is a learned row-grouping classifier to replace SPARTAN's fixed row_tol_frac threshold, which would close its gap on extreme-difficulty tables without sacrificing performance on multi-line wrapped layouts and would directly attack the span loss category that dominates its current error profile. The third is an explicit hybrid model formalised as a routing layer that selects per-region between a structure-focused decoder and an OCR-first extractor based on image_type classification. The infrastructure built for this project, including the dashboard, the parsing layer with per-model CSV export, the evaluation framework, and the per-model runners under a unified interface, is designed to make these next experiments straightforward to run.

6. References

SPARTAN: Heuristic table structure recognition. Scientific Reports (2026). https://www.nature.com/articles/s41598-026-44325-7